Genome-wide association of sub foveal outer segment thickness

Alvin E. Lirio[1]*, Callum Hunt[1], Ha-Jun Yoon[1], Lilian Zevlaris[1], Kayesha Coley[2,3], Chiara Batini[2,3], Mervyn G Thomas[1,4,5]

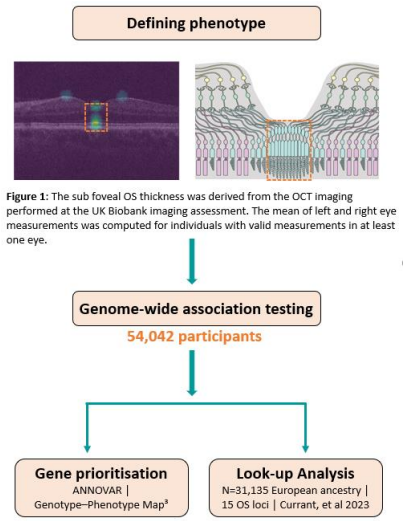
[1]Ulverscroft Eye Unit, School of Psychology and Vision Sciences, University of Leicester, Leicester, United Kingdom; [2]Department of Population Health Sciences, University of Leicester, Leicester, United Kingdom; 3University Hospitals of Leicester NHS Trust, Leicester, United Kingdom; 4Department of Genetics and Genome Biology, University of Leicester, Leicester, United Kingdom; [5]Department of Ophthalmology, University Hospitals of Leicester NHS Trust, Leicester, United Kingdom

Purpose

Why does outer segment (OS) thickness matter?

- The sub foveal OS is the light-capturing compartment of foveal cones. **OS thinning** is an early, measurable marker of photoreceptor damage in AMD and inherited retinal dystrophies, directly correlating with visual acuity loss[1].
- The genetic architecture of sub foveal OS thickness is largely unknown. UK Biobank OCT imaging (>67,000 participants) provides an unprecedented resource for population-scale genetic analysis[2].

Methods



Genome-wide association study

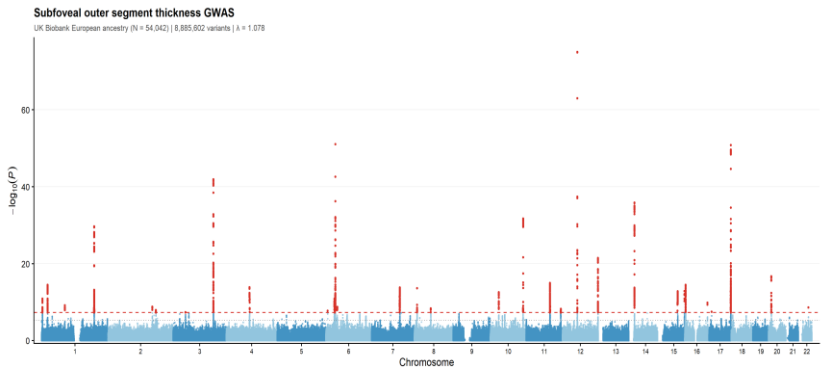


Figure 2: Manhattan Plot. The genome-wide significance threshold is set at $p<5x10^{-8}$ (red dash horizontal line). Associations are plotted as $-\log_{10}(p\text{-value})$ on the y-axis, with chromosome position on the x-axis.

Gene prioritisation: Pleiotropy & functional validation



Figure 3: Functional role of prioritised photoreceptor OS genes.

Look-up analysis

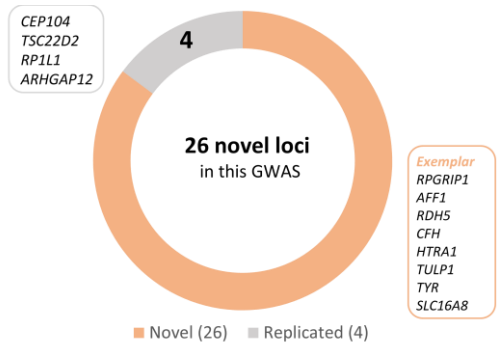


Figure 4: Variant lookup with respect to GWAS signals of all retina photoreceptor thickness.

Conclusion

- This study identifies **30 genome-wide significant loci** for sub foveal OS thickness— **26 novel loci** and **4 replicated** from Currant et al. 2023.
- GP Map colocalisation confirms **3 loci shared with AMD**, links *RDH5* to **myopia** ($P < 10^{-60}$), and reveals pleiotropic associations with other systems including **ciliopathy**, **haematology**, and **immunity**.
- Notably, *RDH5* ($P = 9.3\times10^{-76}$) reached only $P = 5.1\times10^{-4}$ in Currant et al. 2023, demonstrating that **sub foveal-specific phenotyping amplifies foveal cone-specific signals.**
- This work is ongoing to further refine this genetic architecture and prioritise high-confidence candidate genes for downstream investigation.

References

- Currant, H., et al., (2023). PLoS Genet, 19(2), e1010587. https://doi.org/10.1371/journal.pgen.1010587
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al574@leicester.ac.uk | www.linkedin.com/in/alvin-lirio-76a48449